Generalization by Composition: Primitives for Robot Control

INVITED TALK / DARTMOUTH AUTONOMY SEMINARS

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Nov 2026

Intelligence in Motor Skills

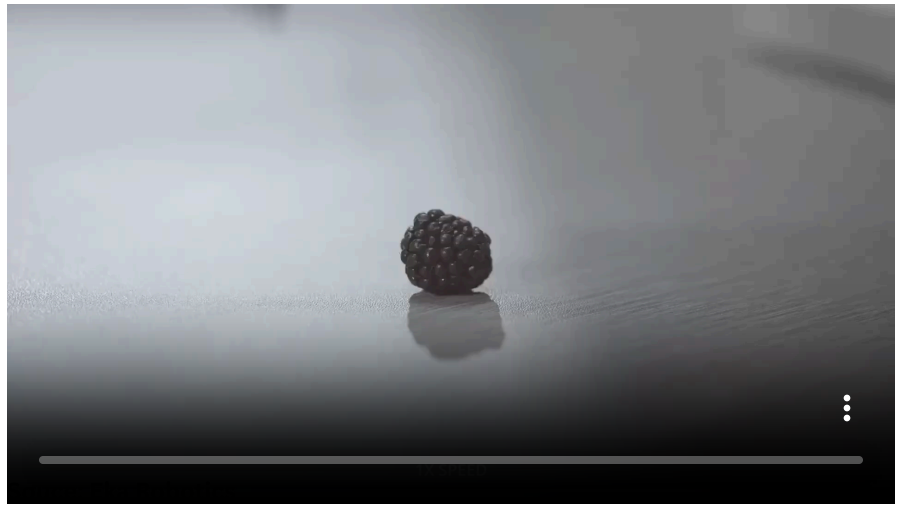
What is intelligence?

- Compression (Shannon, 1951) (Buchanan et al., 2026)
- Prediction (Clark & others, 2013)
- Prediction error (Friston, 2010)
- Embodiment (Brooks, 1991)
- Generalization (Legg & Hutter, 2007)

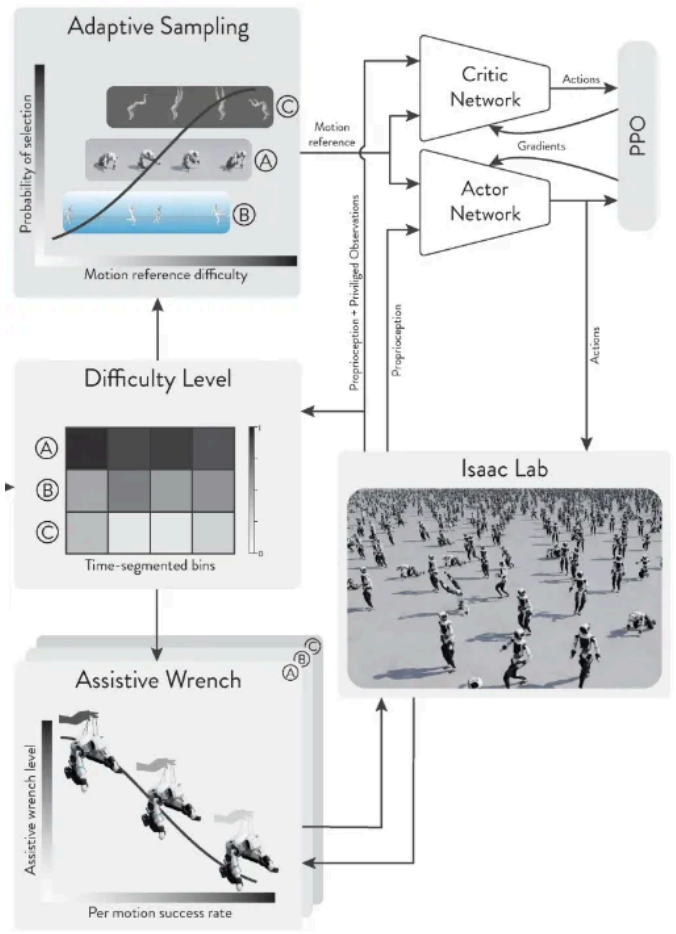
Applications of intelligence

> *I like to talk about intelligence having two parts, an athletic part and a cognitive part.*
>
> *— Marc Raibert*

SoTA performance (EKA-Robotics, 2026) (Sleiman et al., 2026)



RL With Fine-Grained Actions

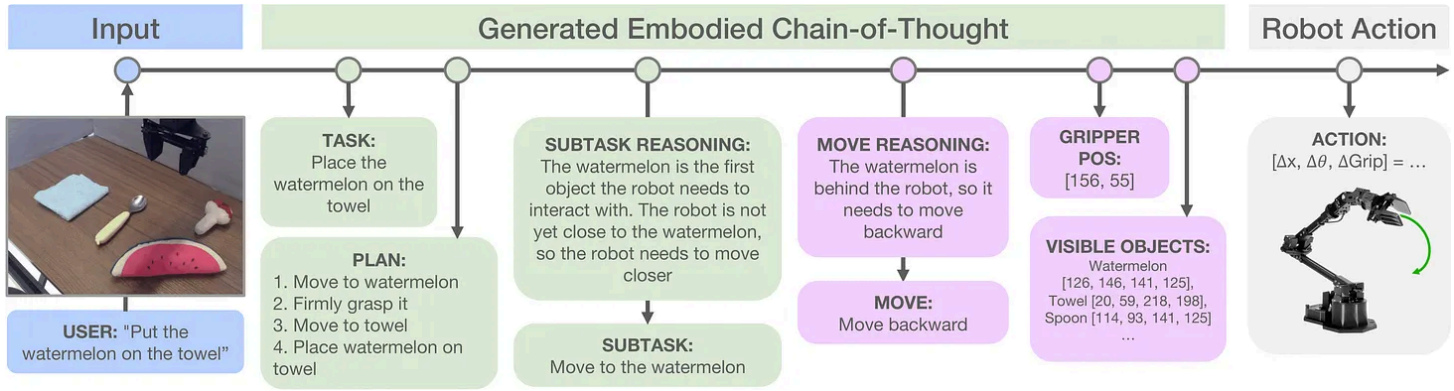


ZEST uses residual joint-space corrections to reference positions with a fixed impedance controller (Sleiman et al., 2026)

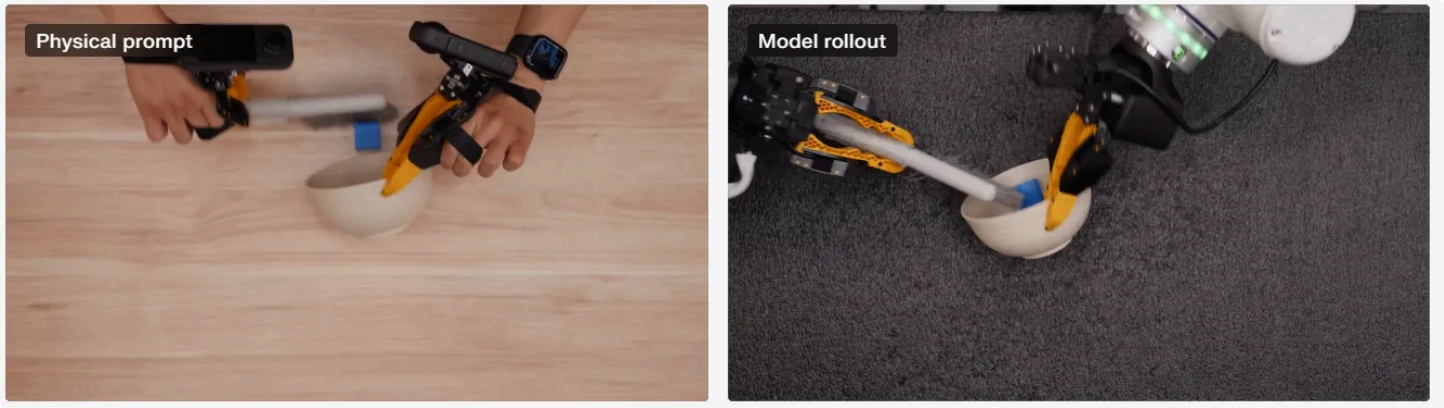
- Using joint actions for RL first showed success in animation (Peng & Van De Panne, 2017)
- Sim2real successes starting with (Haarnoja et al., 2018)
- Fits with the bitter lesson (Sutton, 2019)
- RL's main *challenge* is generalization across **task**, **environment**, and **embodiment** (Ibarz et al., 2021) (Kirk et al., 2023)

RL facilitates flexible, robust, embodied intelligence, but is weak at **generalization**

Task, Embodiment Generalization in Foundation Models



pi0.7 uses a scratchpad for subtask reasoning (Intelligence et al., 2026)



GEN-1.5 in-context learning for generalization to new tasks

Task generalization

Observed:

- Sequential task composition
- Text-based subtask generation
- In-context learning (ICL)

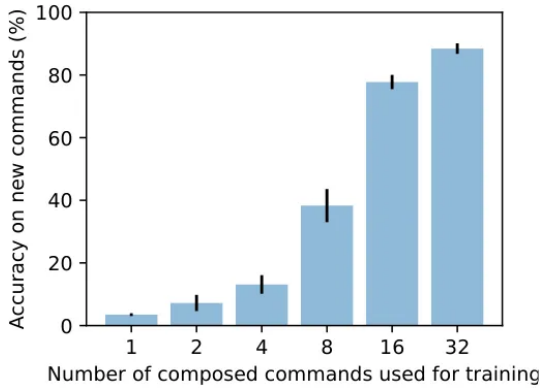
Embodiment generalization

- Tool improvisation
- Large-scale cross-embodiment pretraining*

> *Missing: RL's robustness!*

Extrapolation ("Jump") is Not Solved

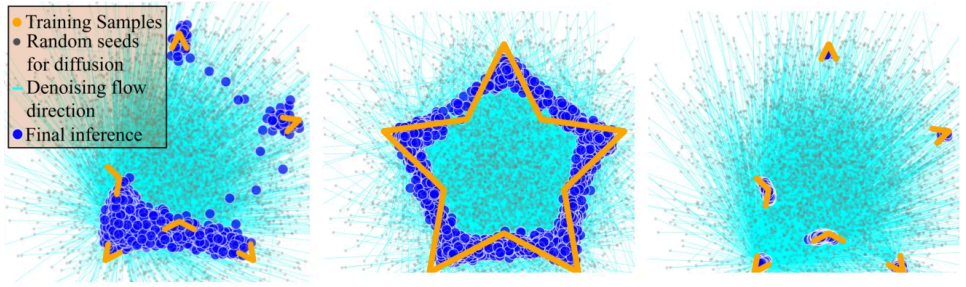
No systematicity



Composed examples needed for combined commands (Lake & Baroni, 2018)

RNNs need training examples in context

Test-time extrapolation



(a) low-capacity model, small data (SD)

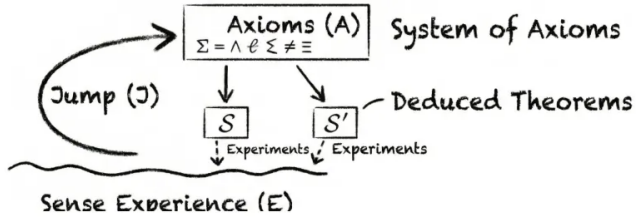
(b) low-capacity model, large data (LD)

(c) high-capacity model, SD (e.g. Diffusion Policy)

Test-time extrapolation of diffusion policies, ACT (He et al., 2026)

Diffusion policies, ACT behave like lookup tables or interpolators, with poor OOD results

LLMs can't jump

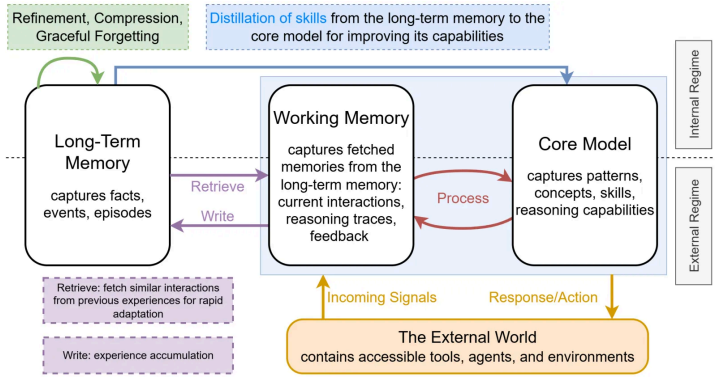


Einstein's Experience → Jump → Action Diagram (Zahavy, 2026)

LLMs cannot form novel hypotheses ("jumps") when there is no error signal gradient for hillclimbing

Adaptation is Not Solved

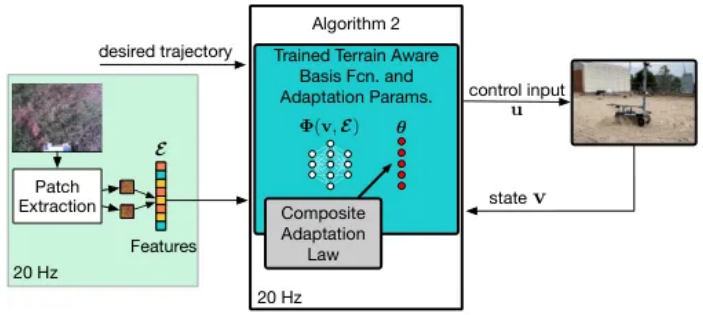
Continual learning



Modular memory architecture (Dorovatas et al., 2026)

We cannot yet continuously acquire new skills without catastrophic forgetting (IWL) or blowing up inference compute (ICL)

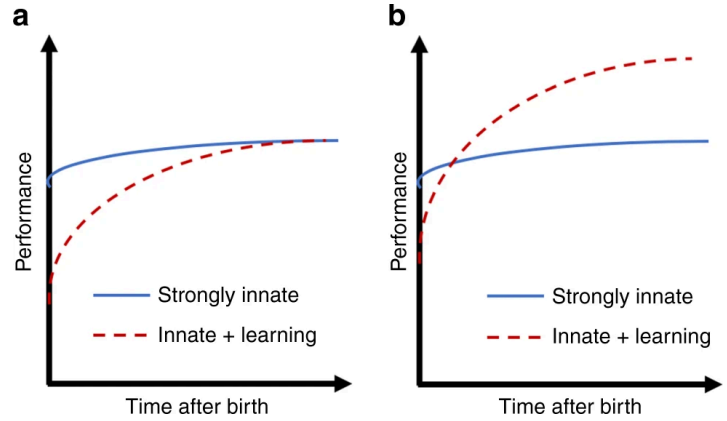
Disturbance rejection



"Last layer" adaptation (Lupu et al., 2024)

- Terrain/environment robustness poor
- Loops and hierarchies acting at different time-scales are key to online adaptation

What Biology does

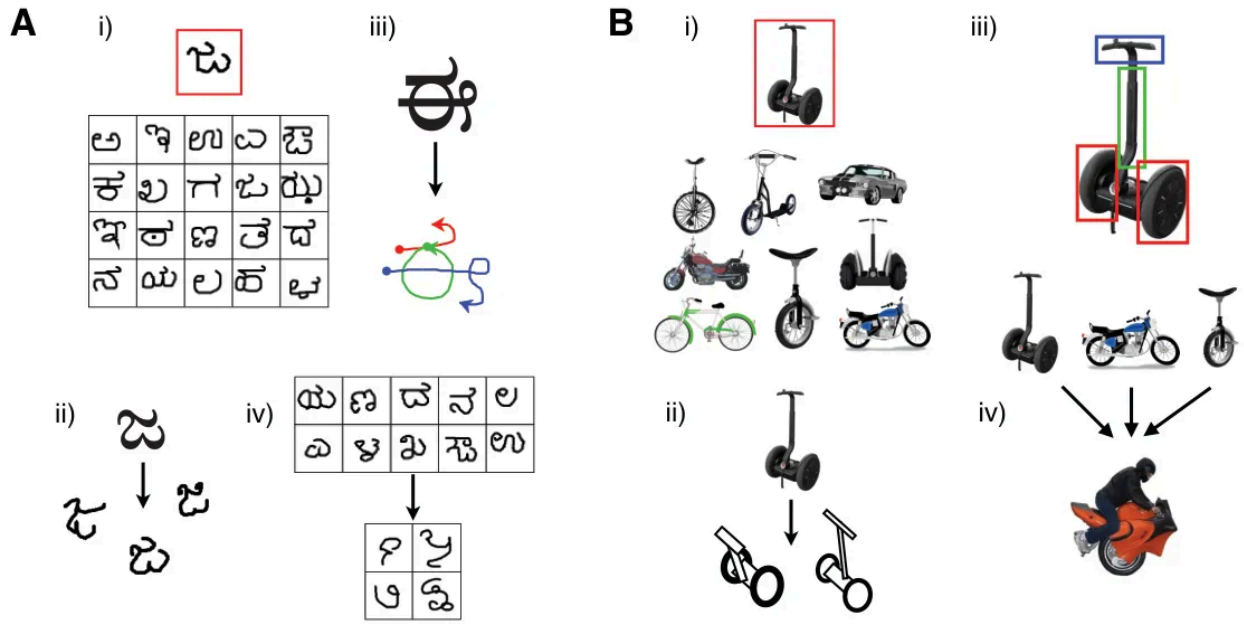


The species relying on a strongly innate strategy will outcompete the species employing a mixed strategy (Zador, 2019)

Animals are able to function well so soon after birth because they are born with highly **structured** brain connectivity

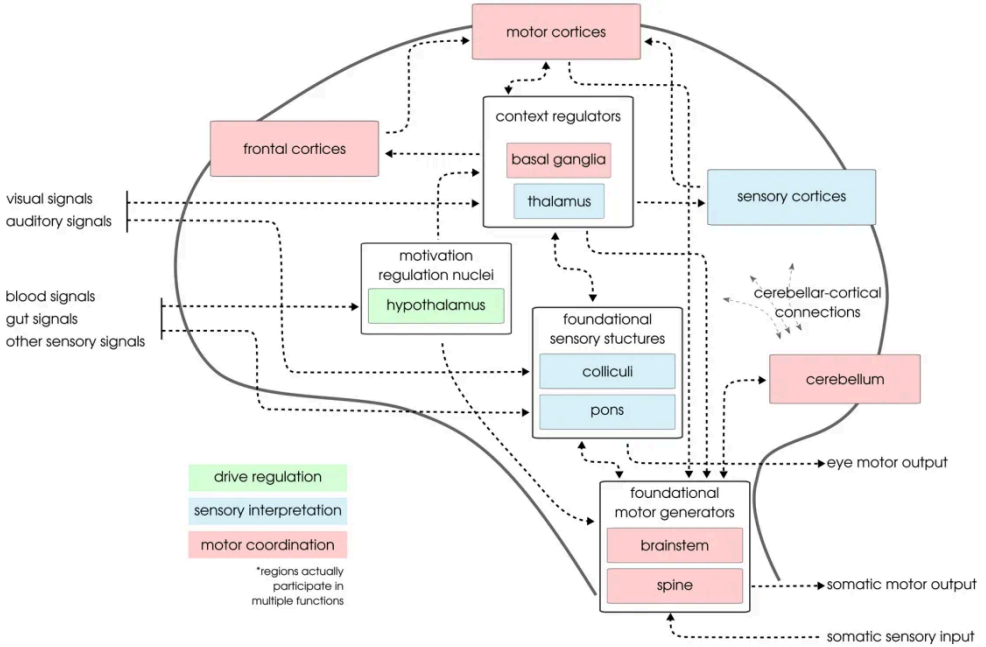
Building Machines That Learn, Think, and Move Like People

What kind of structure?



Pattern-recognition scaling vs. compositional, modular causal models (and physics/motor primitives) for sample-efficient generalization (Lake et al., 2017)

Literally instantiated

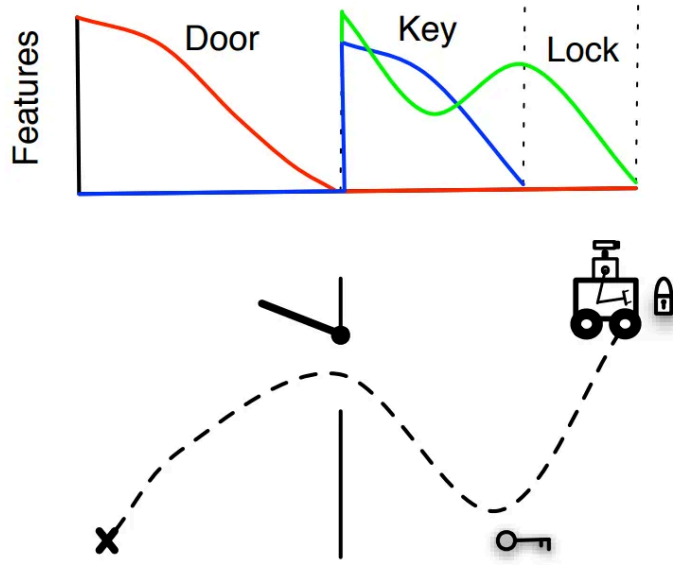


Modular, hierarchical organization of biological motor systems (spinal primitives, brainstem loops, cortical planning) (Merel et al., 2019)

> *Is it because Biology just doesn't have powerful enough computers?*

Computational Challenges are Not Solved

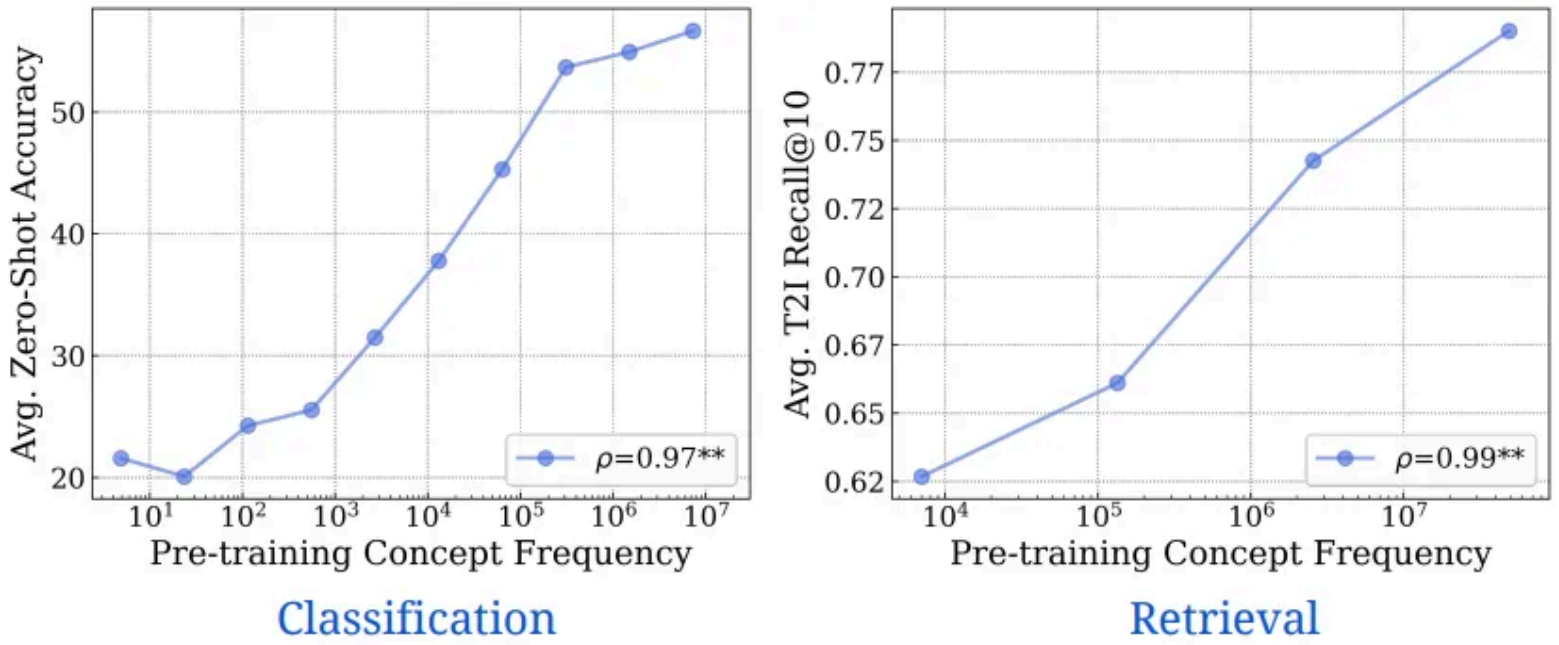
Training complexity



Constructing Skill Trees (Konidaris et al., 2012)

$O(N_{\text{skills}})$ vs. $O(N_{\text{sequences}})$ demonstrations with explicit composition

Exponential scaling



Linear improvements in downstream "zero-shot" capability require exponentially more pretraining data on those concepts (Udandarao et al., 2024)

Outline

Thesis

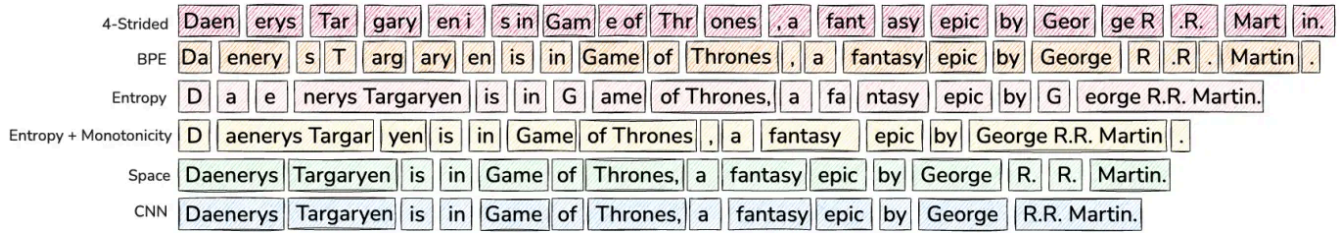
> *An intelligence architecture should be **generative, adaptive, efficient, and embodied** — able to understand and use the body's dynamics, synthesize novel behaviors, and efficiently adapt to new tasks and environments.*

Organization

1. **Introduction**
2. The Case For Dynamical Action Primitives
3. Building Composable Action Primitives
4. Open Questions

The Case For Dynamical Action Primitives

An Analogy: Tokens in LLMs



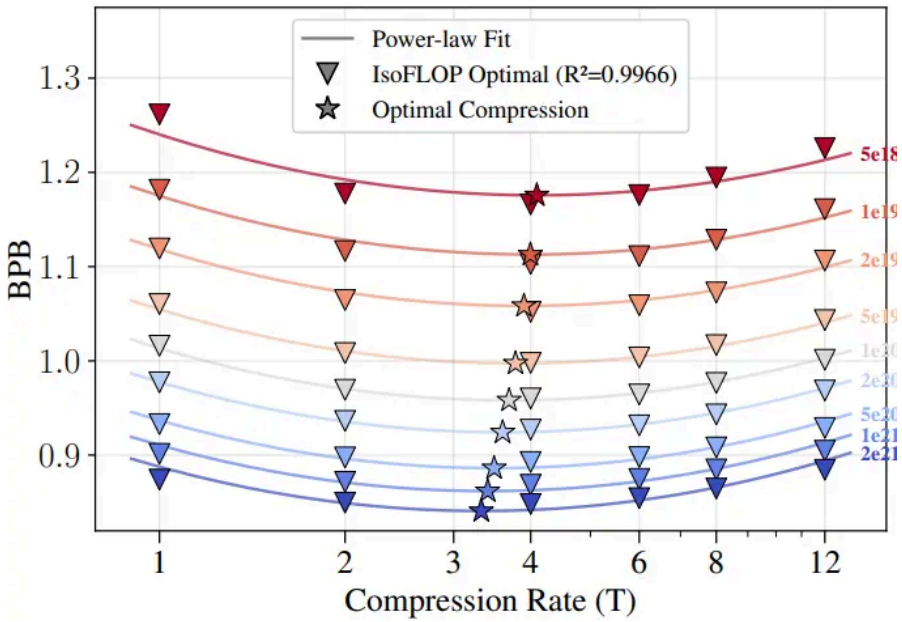
Strategies for grouping bytes into patches (Pagnoni et al., 2025)

Vocabulary size / token size tradeoff

Letters (fine) vs. words (coarse)

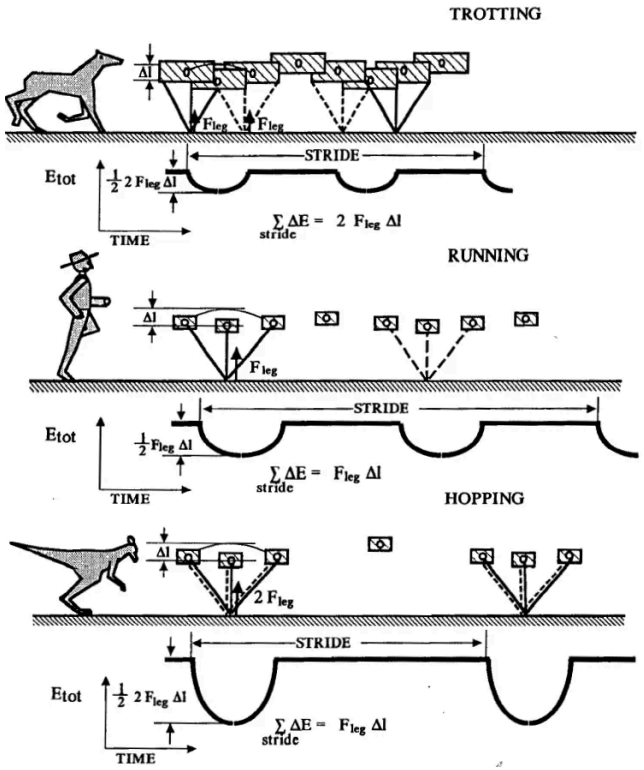
- Memory: higher for coarse tokens
- Computation: longer sequences for fine tokens
- Generation: sequences of words vs. letters
- Transfer: coarse are task-specific, but transfer better

E.g. medical journal, English-Italian dictionary

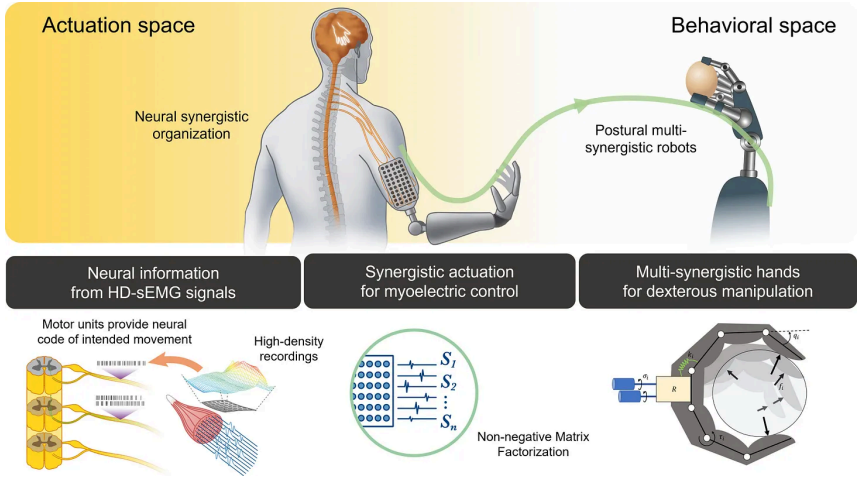


Loss (BPB) vs. bytes/token (compression rate) shows an optimum at each training budget (Limisiewicz et al., 2026)

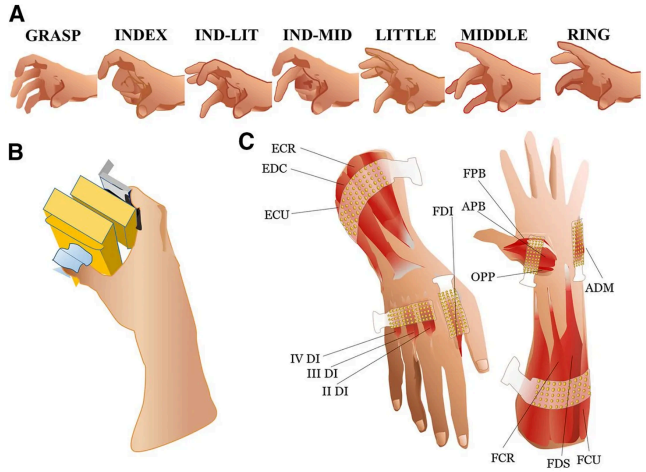
Animals Exhibit Coarse Action Primitives



Bouncing like a monopode (Blickhan & Full, 1993)

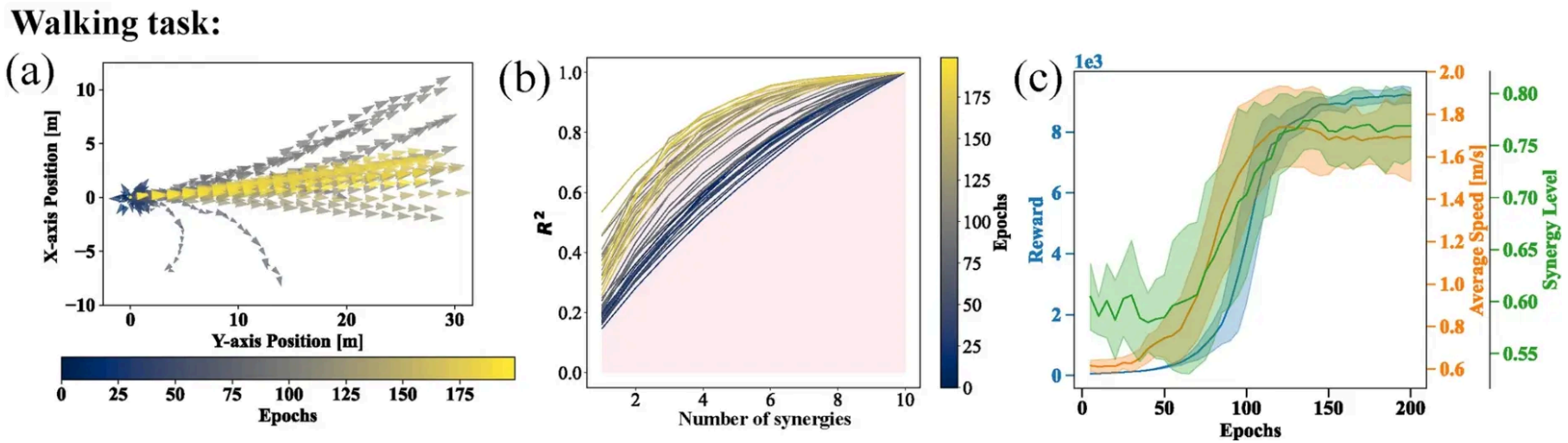


Neural decoding of synergistic behavior of spinal motoneurons (Capsi-Morales et al., 2025)

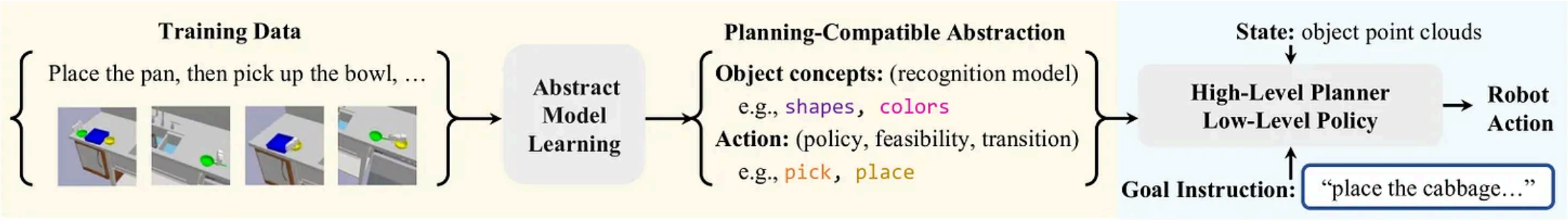


Motor neuron activities exhibit a synergistic organization in complex tasks; large dimensionality reduction in motor neuron outputs (Tanzarella et al., 2021)

Learned Models Exhibit Coarse Action Primitives



With only energy and the motor performance in the reward, synergies (via PCA on joint torques) emerge (Li & Hayashibe, 2025)



Learning planning abstractions in sequential decision-making domains from language-annotated demonstrations (Liu et al., 2024)

What We Want: Dynamical Primitives

Fish with no active control (Beal et al., 2006)

Embrace dynamics

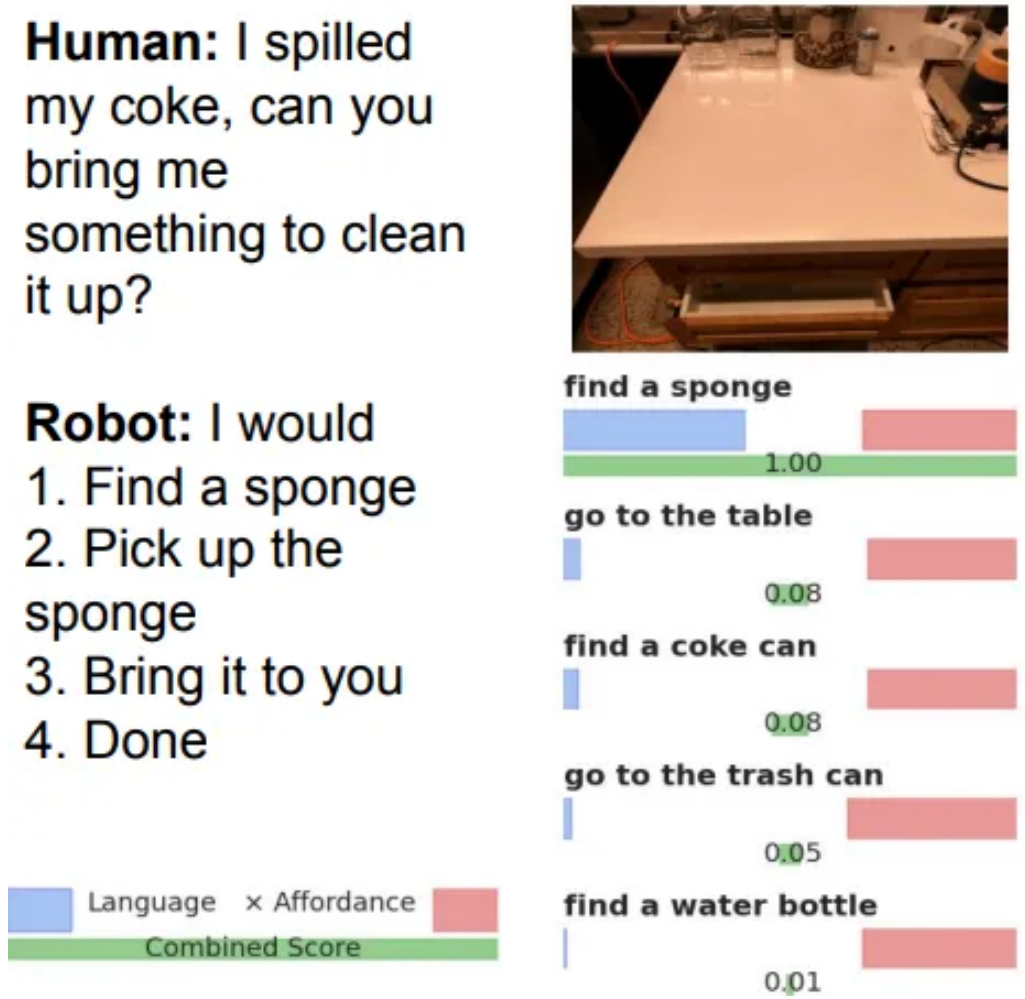
- Kangaroo hopping vs. human running need different strategies
- Unconventional robot designs
- Highly dynamic behaviors like gymnastics
- Mechanical intelligence is invisible when training from video data

Dynamical stability and robustness

- Adapt to terrain disturbances (Lupu et al., 2024)
- The “ICL” equivalent for physical work

> *Cannot learn from imitation/pre-training!*

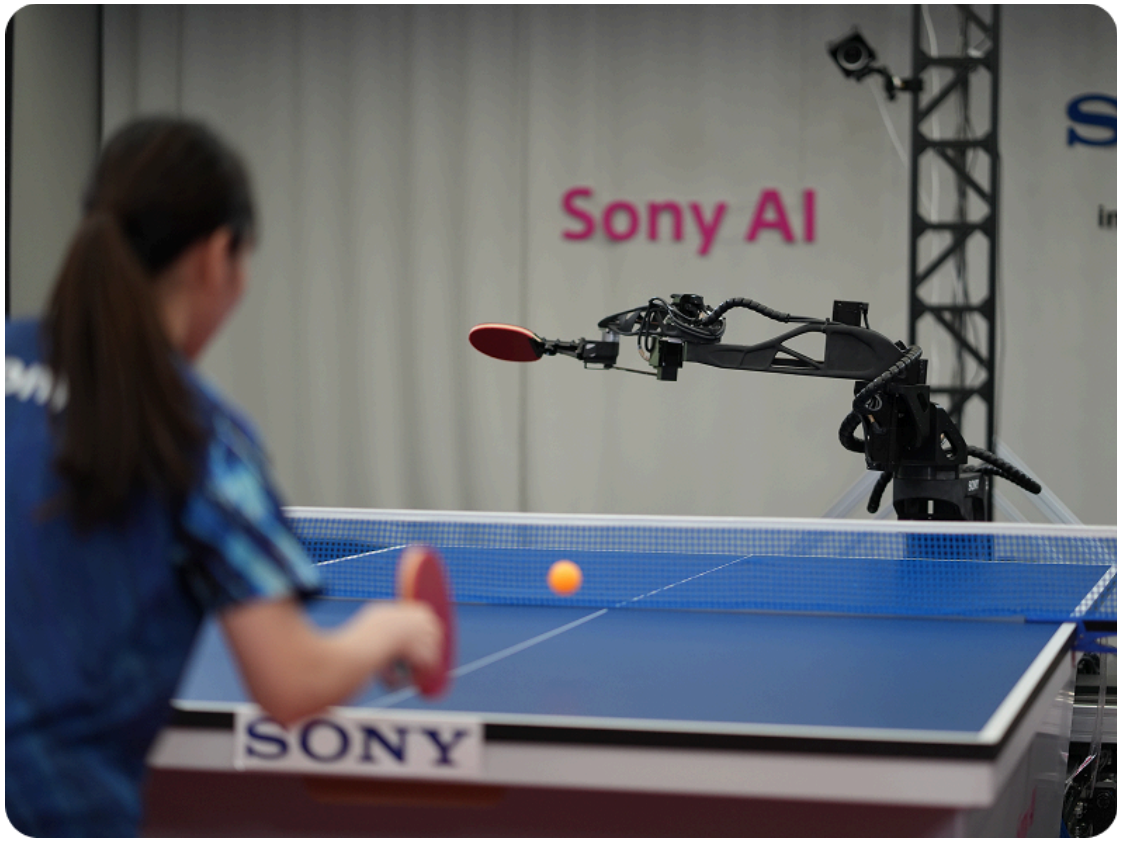
What We Want: Explicit Sequential Composability



- "Emergent" language-driven sequencing → explicit, guaranteed, and dynamical sequential composition
- Dynamics pose correctness challenges: e.g. `walk` → `long jump` not feasible; `walk` → `run` → `long jump` is

SayCan (Ahn et al., 2022), similar to pi0.7 (Intelligence et al., 2026)

What We Want: Explicit Parallel Composability



Sony Ace elite-level table tennis robot (Durr et al., 2026).

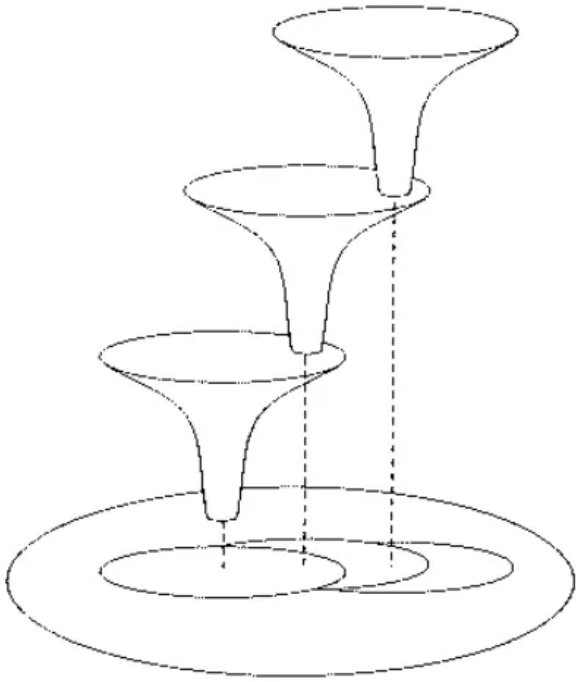
- E.g. combine upper and lower-body skills
- Poses similar dynamic stability challenges

Building Composable Action Primitives

Formal Sequential Composition using Dynamical Models

"Buegler" robot palming, catching, juggling

(Burridge et al., 1999)



- Formal guarantee of composability
- Idea: each controller's goal state inside domain of attraction of next

Empirical Dynamical Parallel Composition

"Control in 3 parts" (Raibert, 1986)

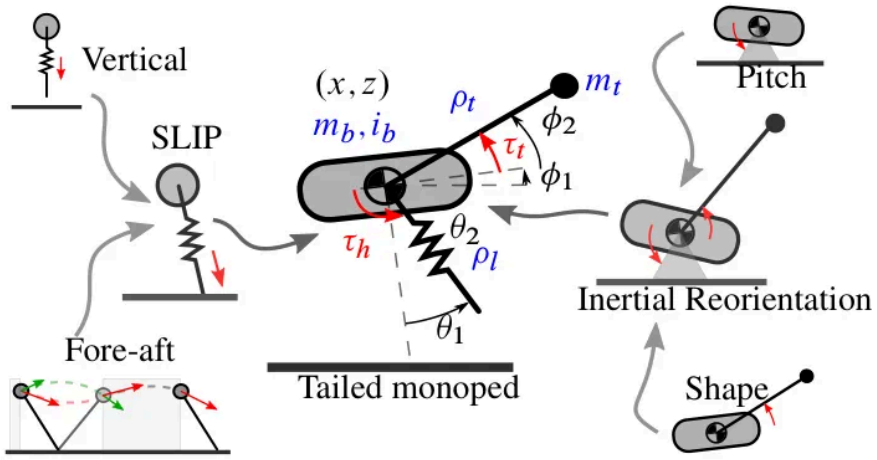
- Control of hopping height
- Control of forward speed
- Control body attitude

Control of Running Decomposed into Three Parts

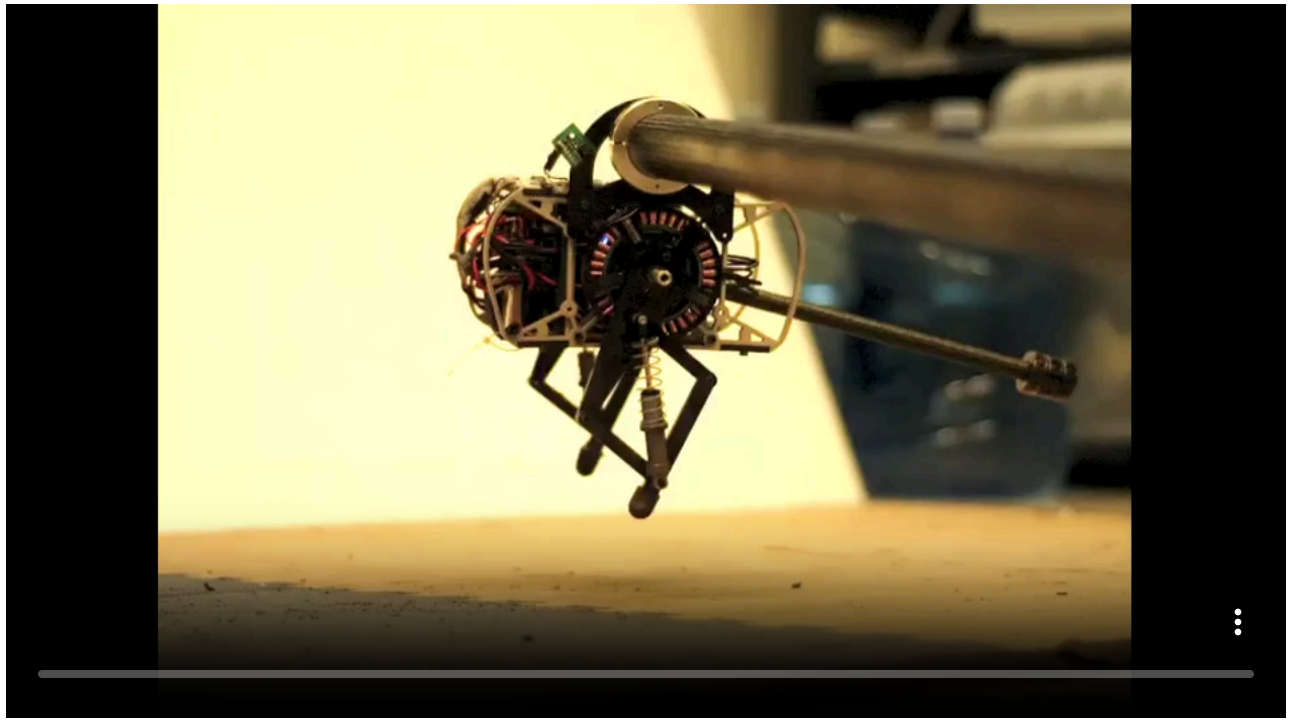
The control system we explored for the planar one-legged machine treats hopping, forward speed, and body attitude as three separate control problems. One part of the control excites the hopping motion and regulates

Specific to a particular type of design – would not work with a high-inertia leg.

Formalized Parallel Composition for Tailed Hopping



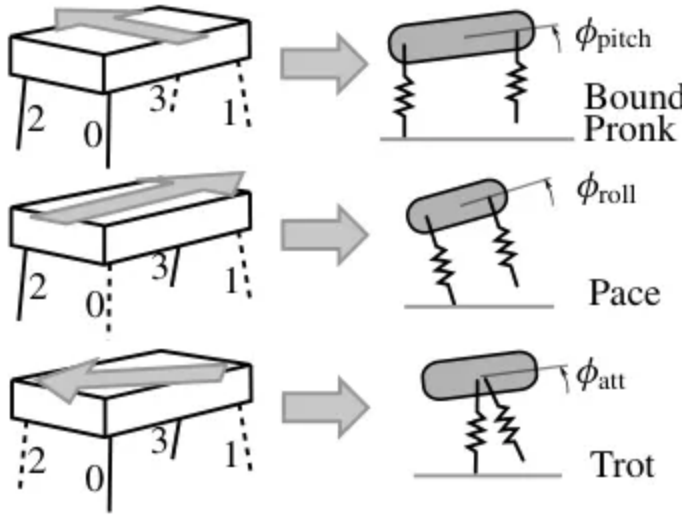
(De & Koditschek, 2015) (Shamsah et al., 2018)



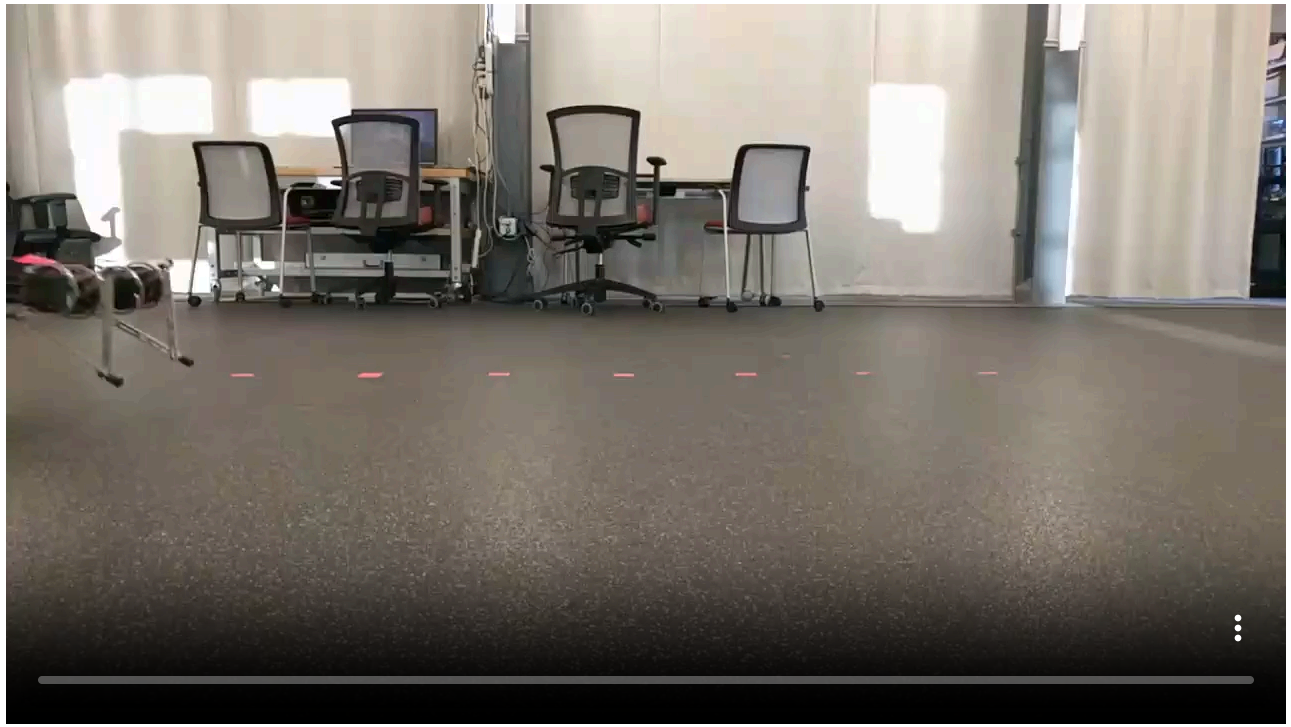
Template-based control

- Projection: $p_T = \Pi(p)$
- Template dynamics: $\dot{x}_T = f(x_T, u_T; p_T)$
- Anchoring (controller embedding): $u = \Pi^{\dagger}(u_T)$

Minitaur Vertical Hopping Compositions



Virtual biped permutations (De & Koditschek, 2018)



- Decoupled, decentralized control + mechanical coordination
- **Embodiment transfer** between Minitaur/Jerboa (different projection & embedding)
- **Task transfer** (generate different gaits)

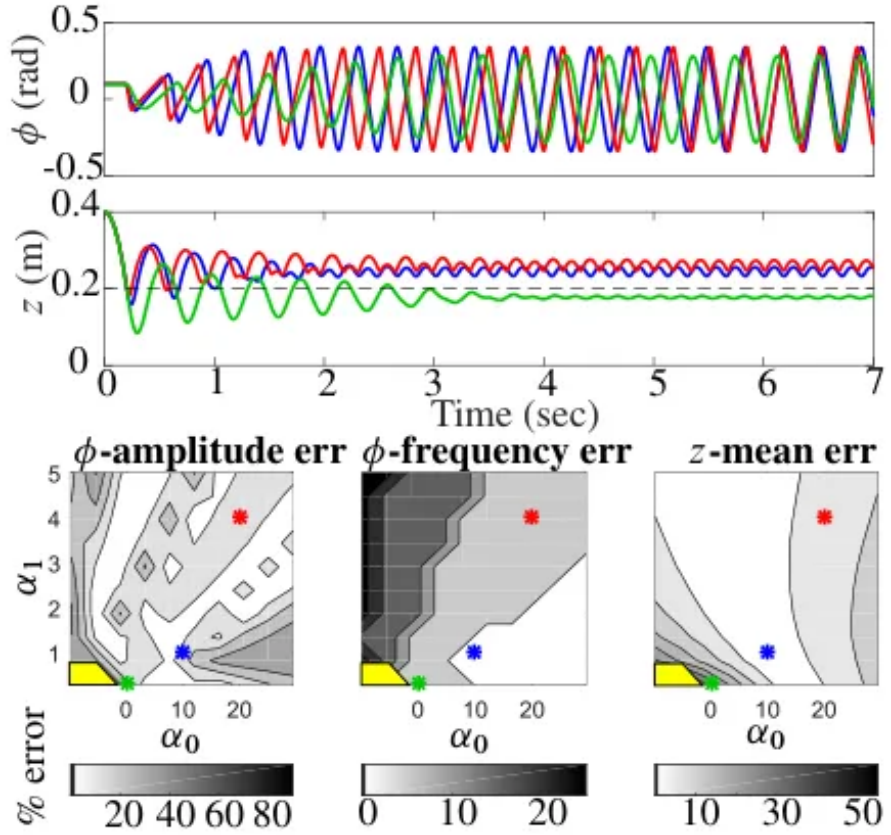
Minitaur Vertical Hopping Model-Robustness

Robust to modeling errors

- α non-dimensionalized parameters varied $\sim 10\times$
- Similar to RL with aggressive domain randomization
- **Weakness of model-based optimal control methods**

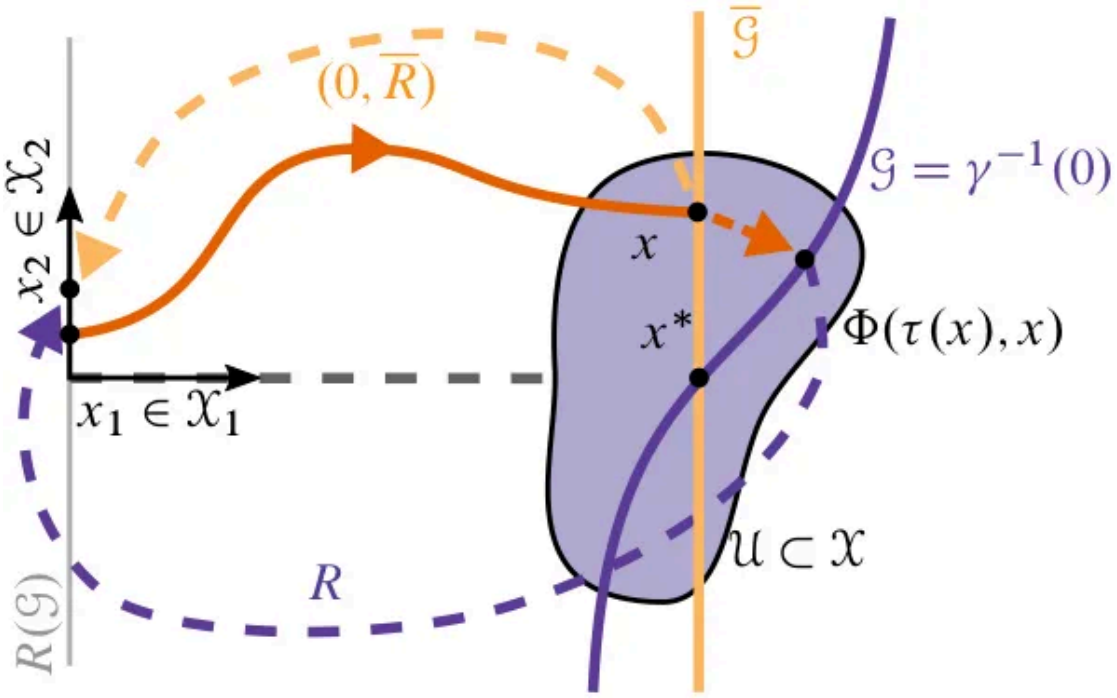
Why does it work?

- Simplicity of f and the (lack of) sensitivity of the Π function



Parametric robustness (De & Koditschek, 2018)

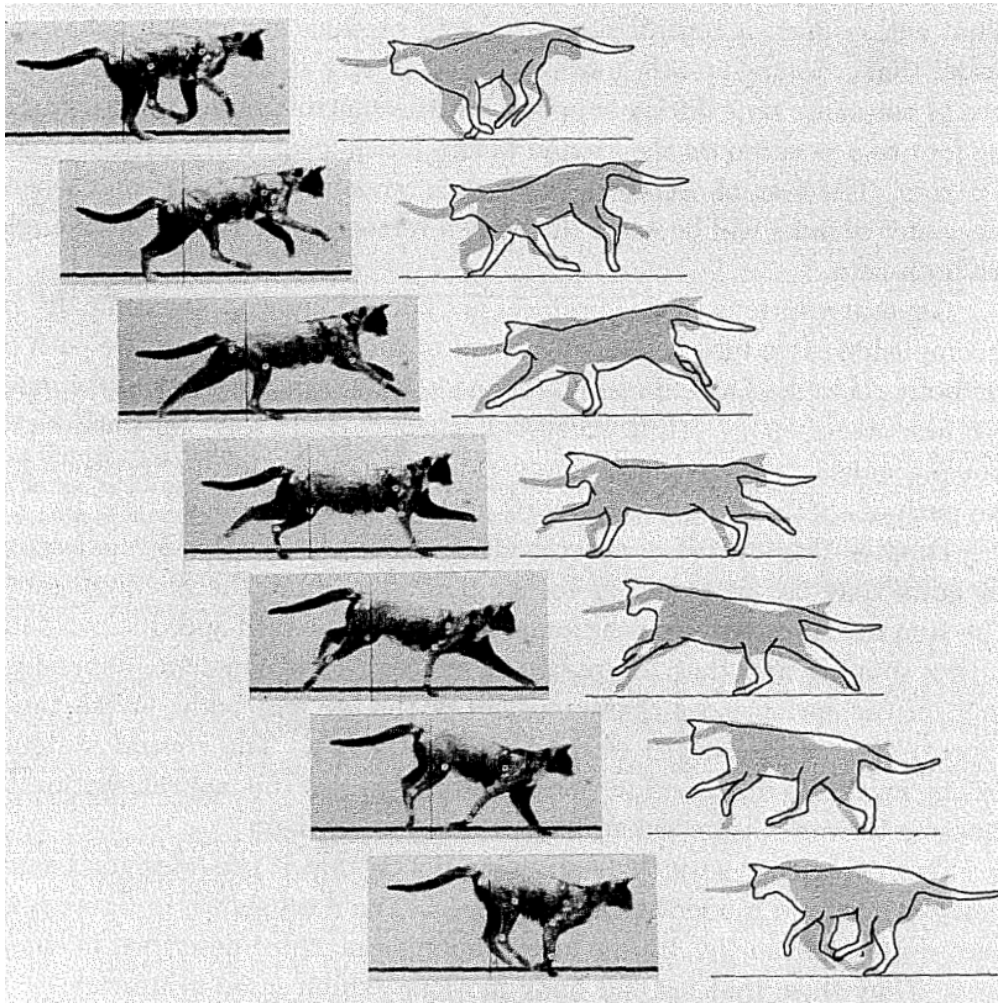
“Correct By Design”: Hybrid Averaging



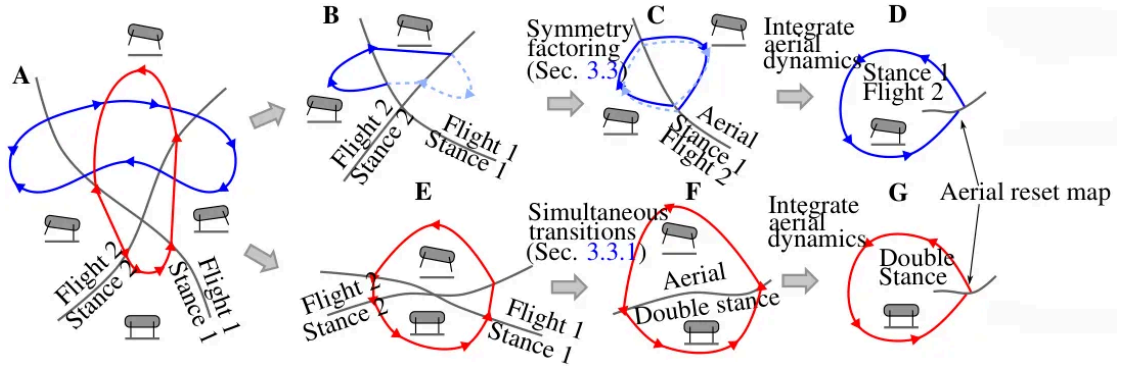
Hybrid averaging theory to find asymptotic behavior of composed cyclic systems
(De et al., 2018)

- **Analytically:** Reduce complex behaviors to simple ones
- **Synthetically:** Allows empirical composition to work

“Correct By Design”: Time-Reversal Symmetry

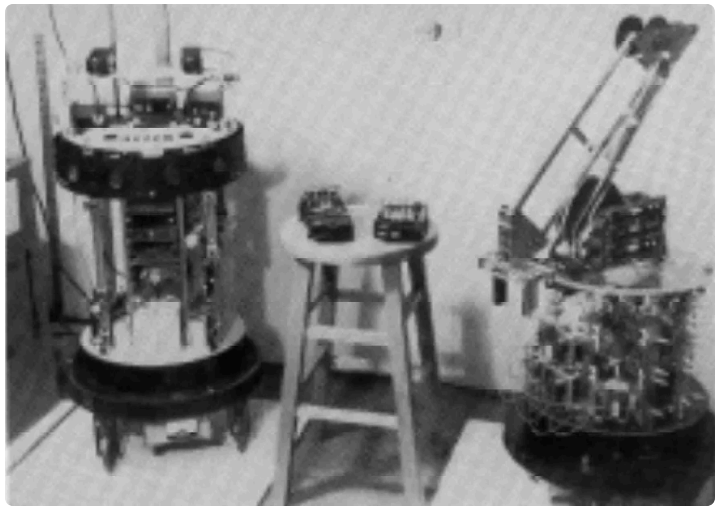


Gallopping cat image from (Raibert, 1986)

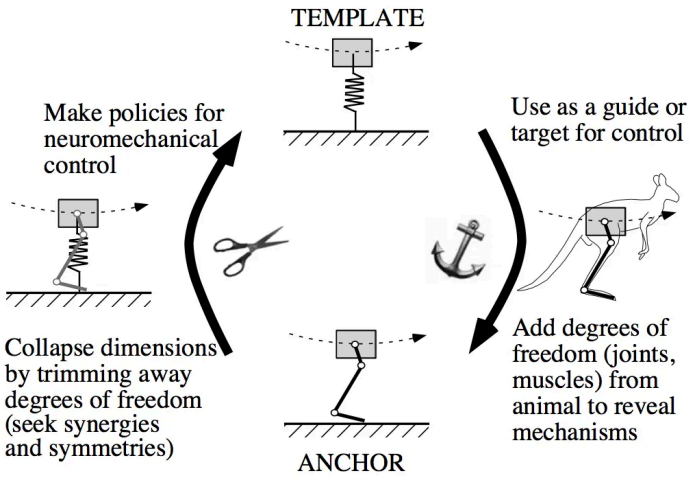


Dimension reduction using time-reversal symmetry complements averaging.

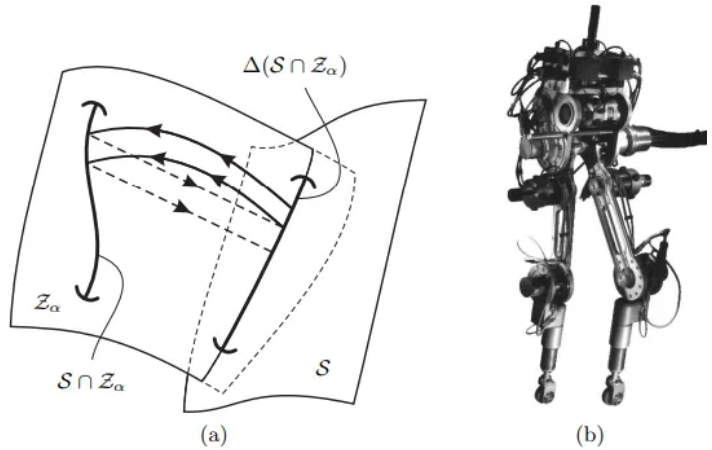
Hierarchical Control Modularity



Subsumption architecture (Brooks, 1991)



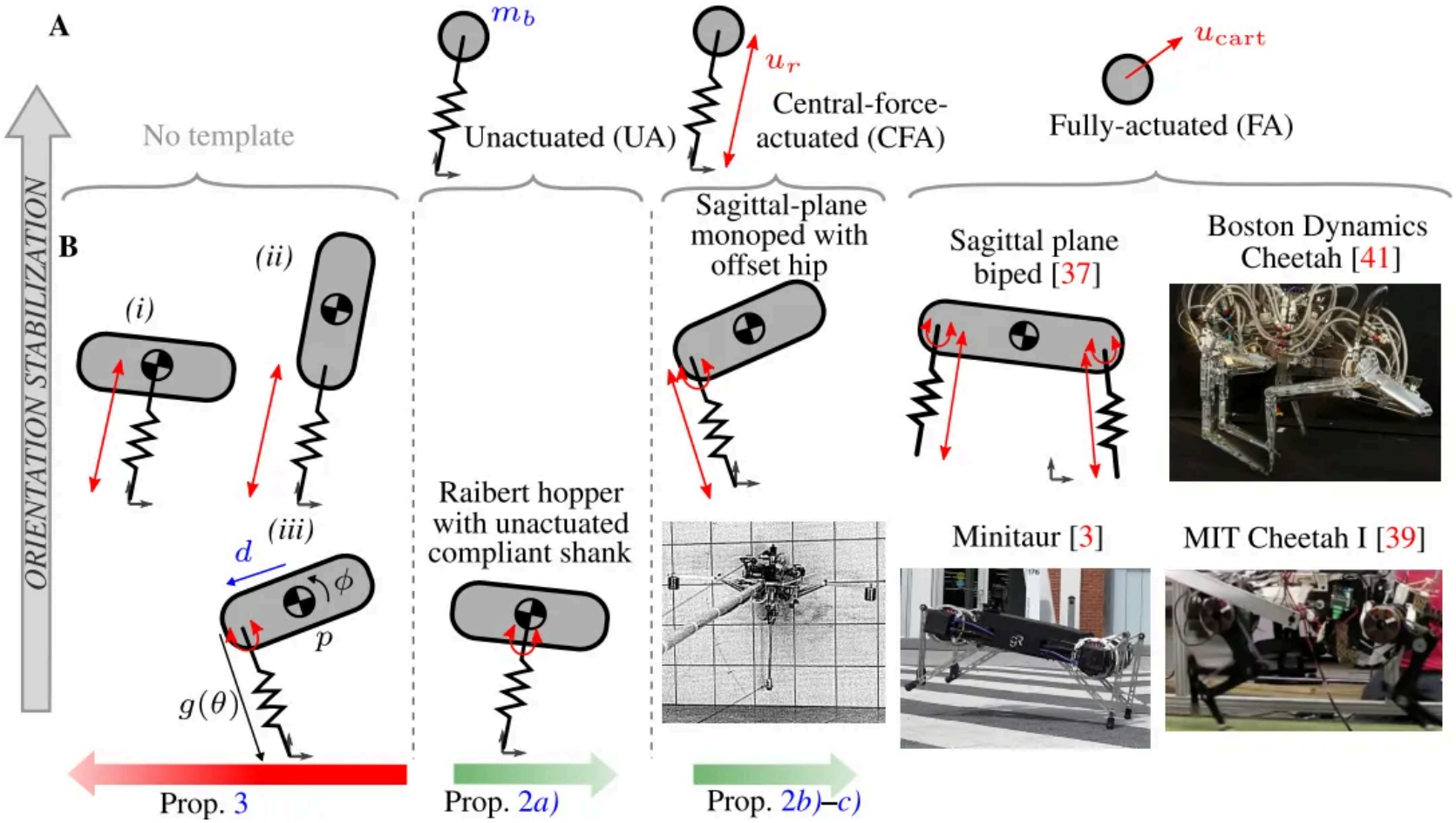
Template and anchors to describe the behavior of animals and robots (Full & Koditschek, 1999)



Hybrid Zero Dynamics formalism of dimension reduction with posture control (Westervelt et al., 2003).

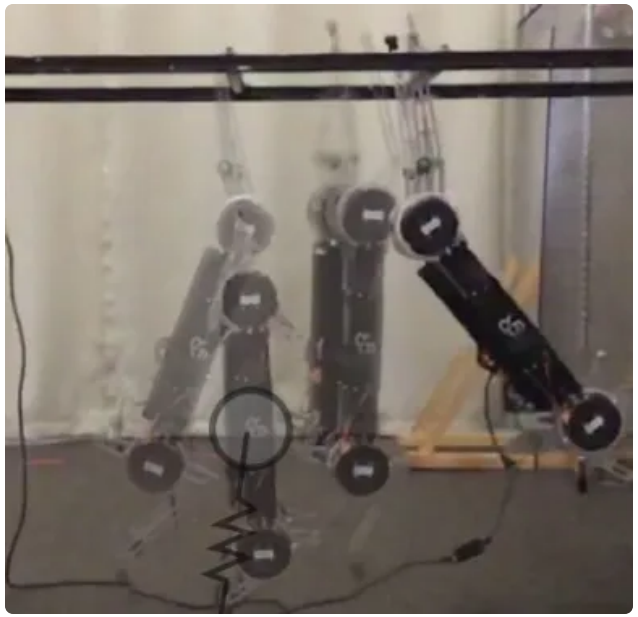
- Alternative to a `Sense` → `Plan` → `Act` , or a linear VLA architecture
- Explicit hierarchies facilitate multi-timescale control, embodiment transfer

Center-of-Mass (CoM) Primitives Guaranteed to Appear

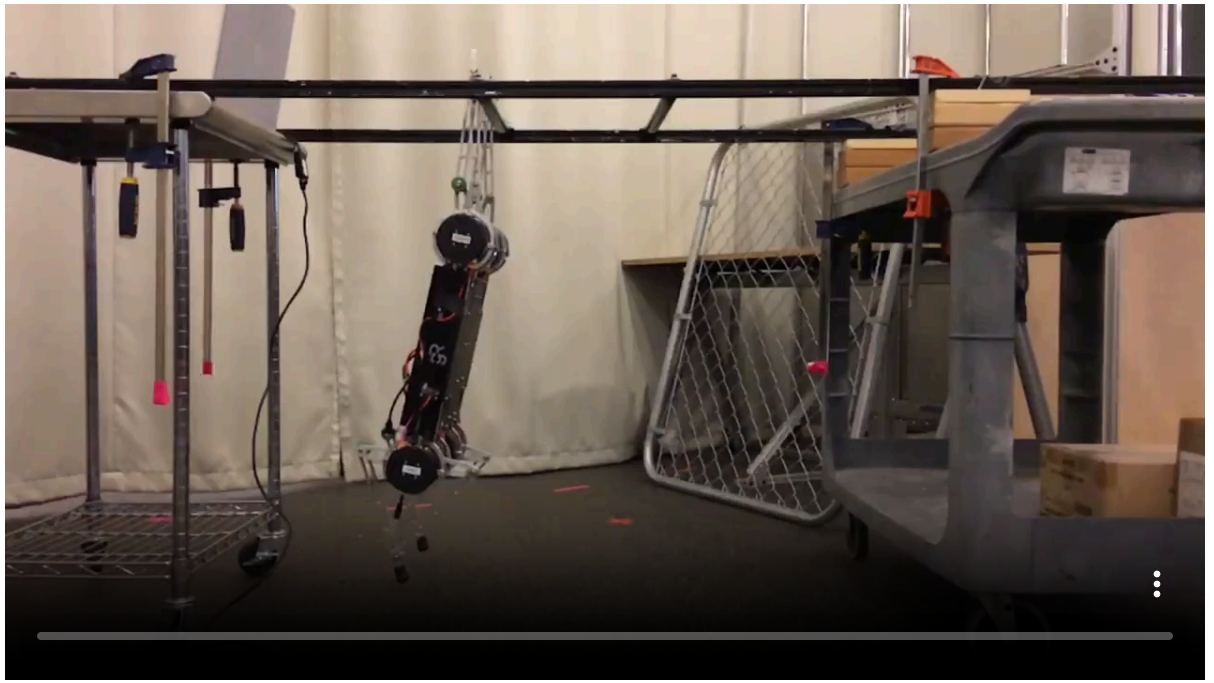


In low-dimensional sagittal-plane models, posture control results in necessary dimension reduction to a specific form of CoM dynamics (De et al., 2022)

Embed CoM Primitive Task Controller



Mode-reactive templates (De et al., 2022)



```
Analysis:  Full model       →    postural dimension reduction    →       CoM dynamics
Synthesis: Robot control    ←           posture control          ←   CoM task control
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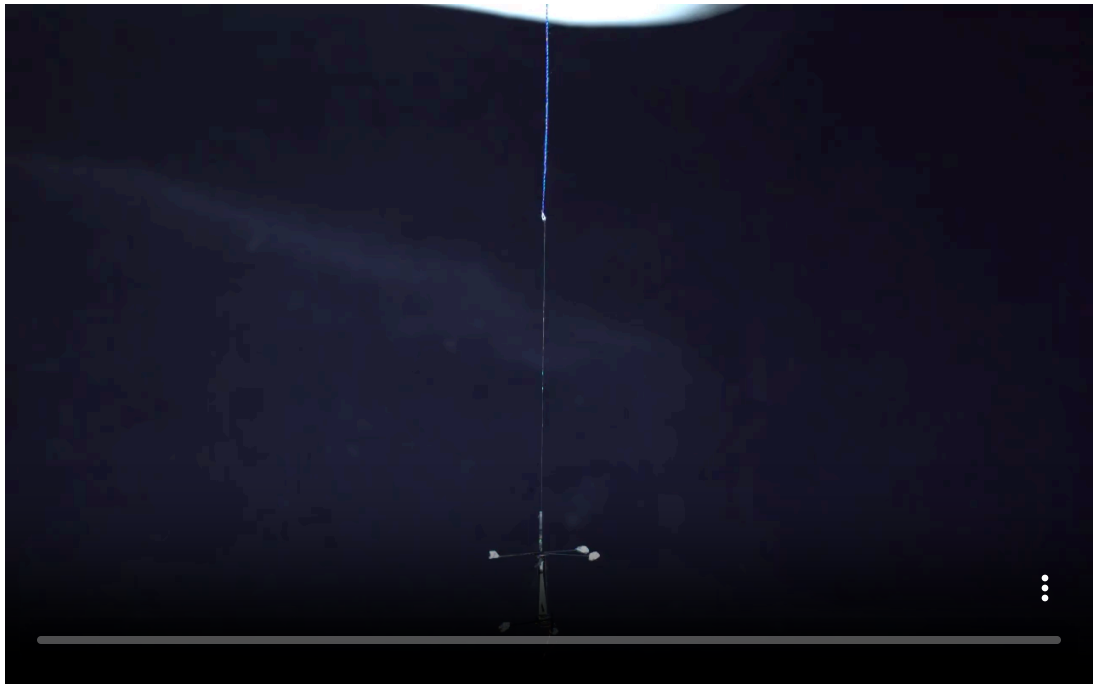
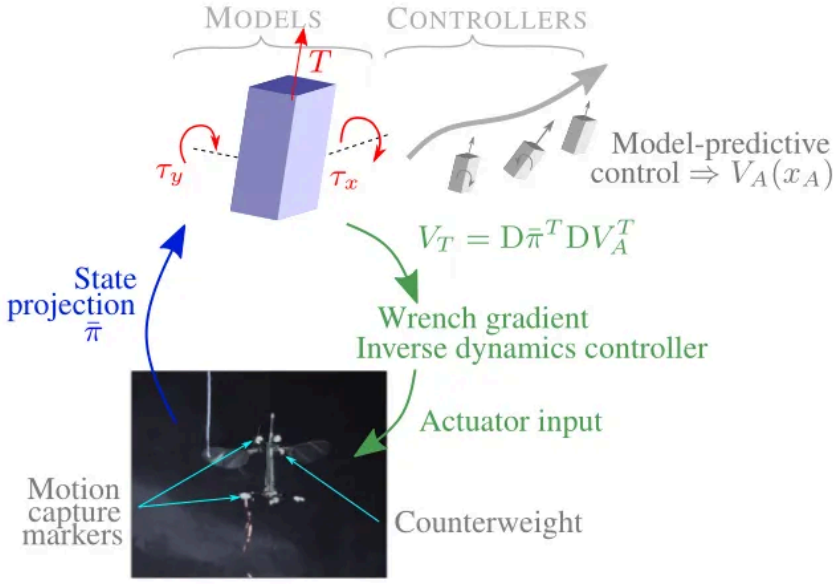
> *"Automated" Π function in certain settings*

Inverse Dynamics Action Primitive: MPC for RoboBee

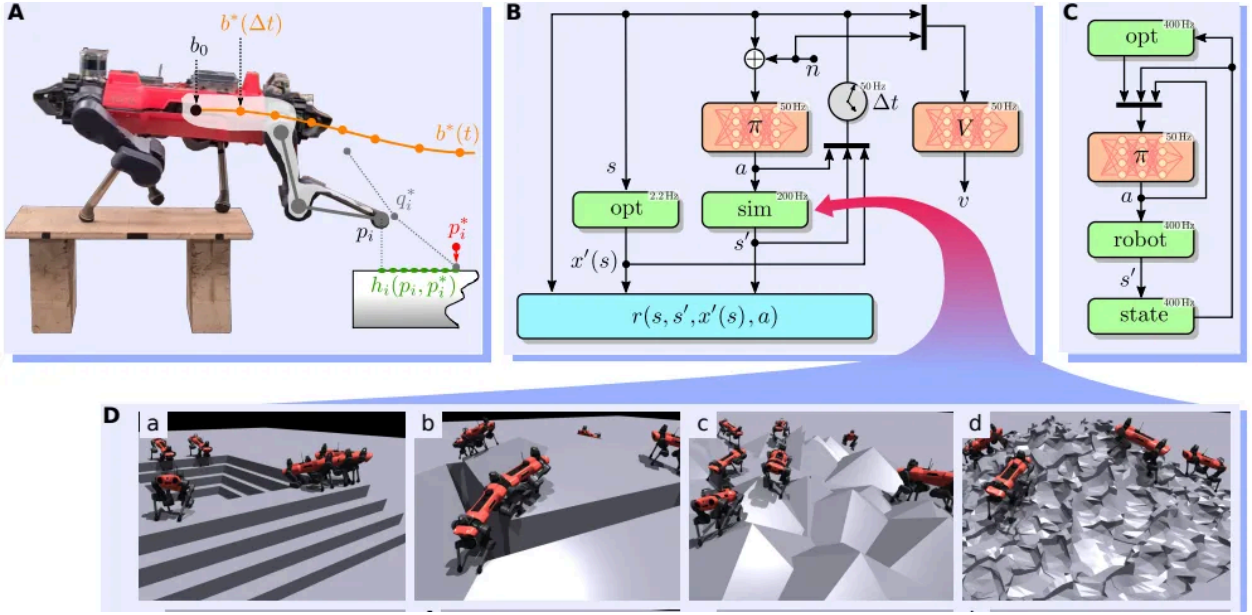




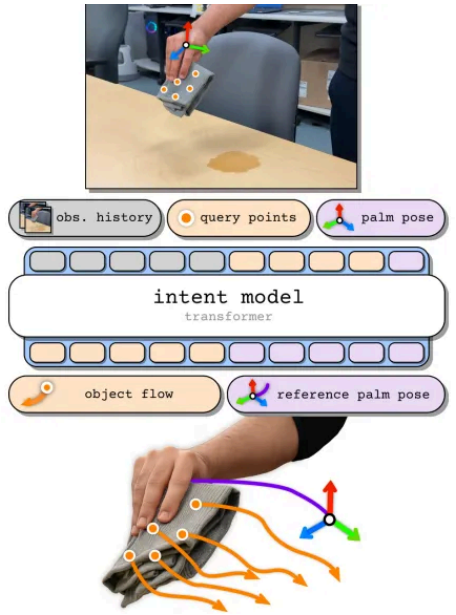
Hierarchical MPC for RoboBee (De et al., 2022)

- MPC (higher-level): online generalization
- Inverse Dynamics (lower-level): model-robustness, embodiment transfer

RL Action Primitive (Fixed Interface)



Traj opt → RL tracking – Deep Tracking control (DTC) traj opt → RL tracking (Jenelten et al., 2024)

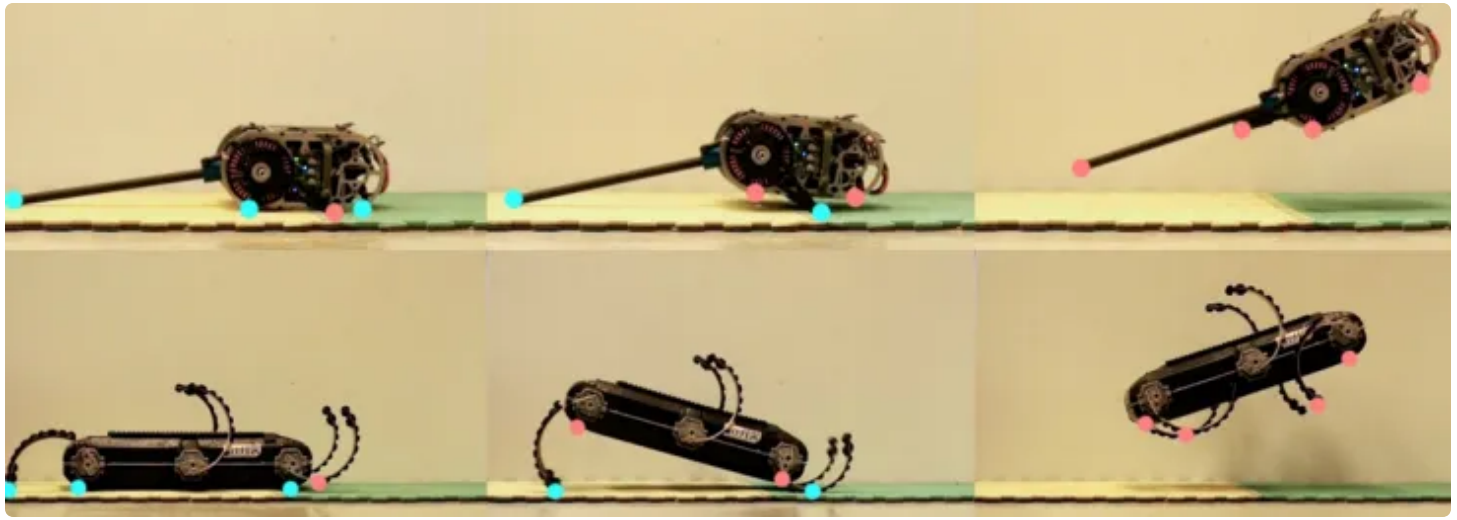


Intent model → RL tracking – LUCID (Gupta et al., 2026)

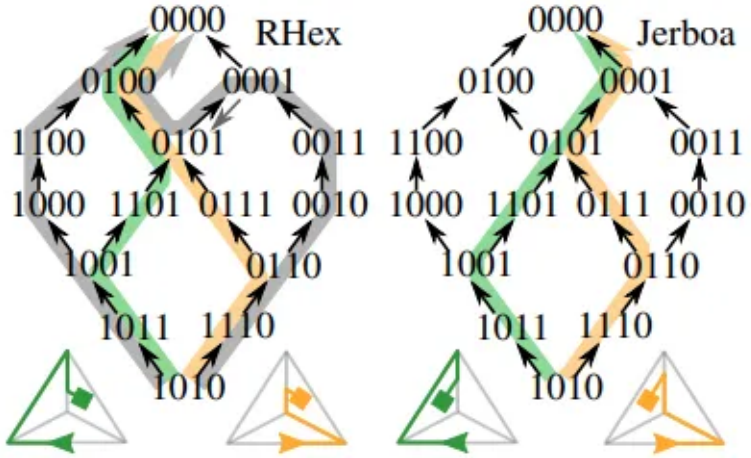
- RL can allow for faster synthesis of robust primitives when sufficiently specified
- Dynamical, embodied, robust, though not guaranteed to be composable

Symbolic Planning for Leaping

- Contact-heavy planning with torque control ⟶ difficult online (Posa et al., 2014)
- Characterize “letters” ⟶ cheap online planning



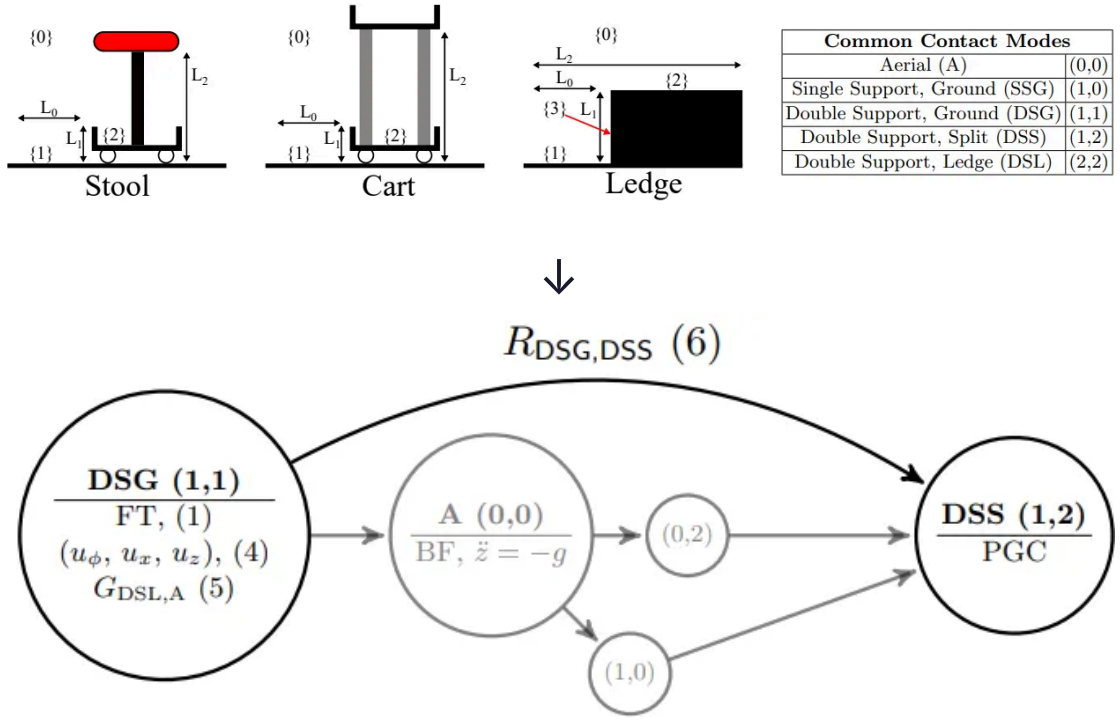
Leaping behaviors built by composing sequences of hardcoded transitions (Brill et al., 2015)

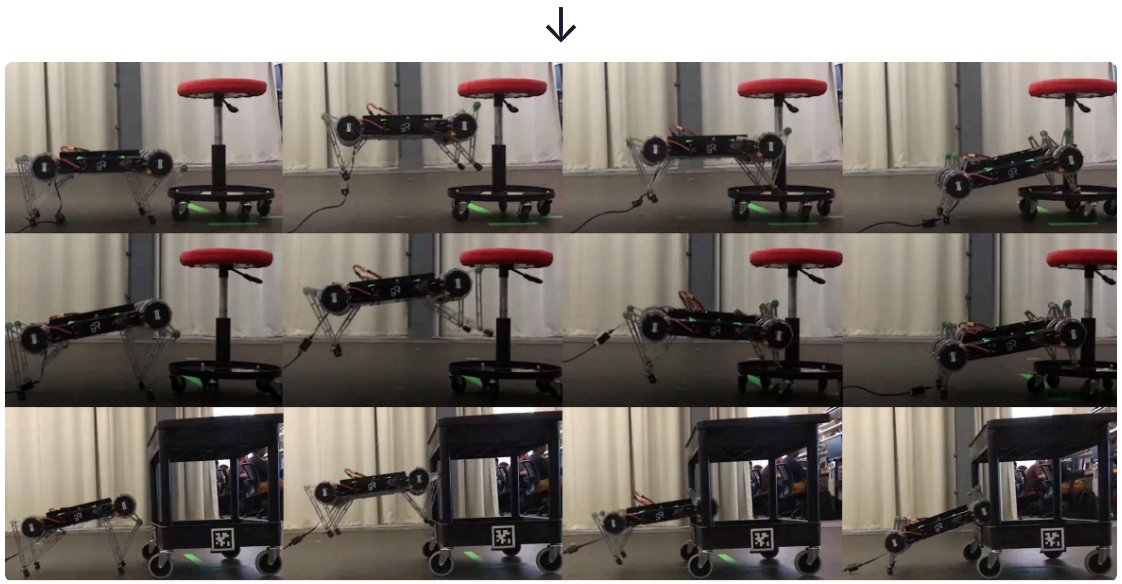


Ground reaction complex (GRC)

Symbolic Planning for Manipulation

- Contact-heavy planning with torque control ⟶ difficult online (Mordatch et al., 2012)
- Characterize “letters” ⟶ cheap online planning





(Topping et al., 2019)

What Have We Built So Far?

Action primitives that are...

Dynamic

Hopping, leaping, running, etc.; utilizing passive dynamics

Generative

Online recombination sequentially, parallelly

Efficient

Traj opt → cheap online discrete symbol-space planning

Transferable

Robust to modeling errors; swappable body params

We (still) need an architecture capable of synthesizing, as well as autonomously tasking and composing primitives

Open Questions

Best of Both Worlds: Semantic FM + Dynamical Primitives

Foundation model (high-level)

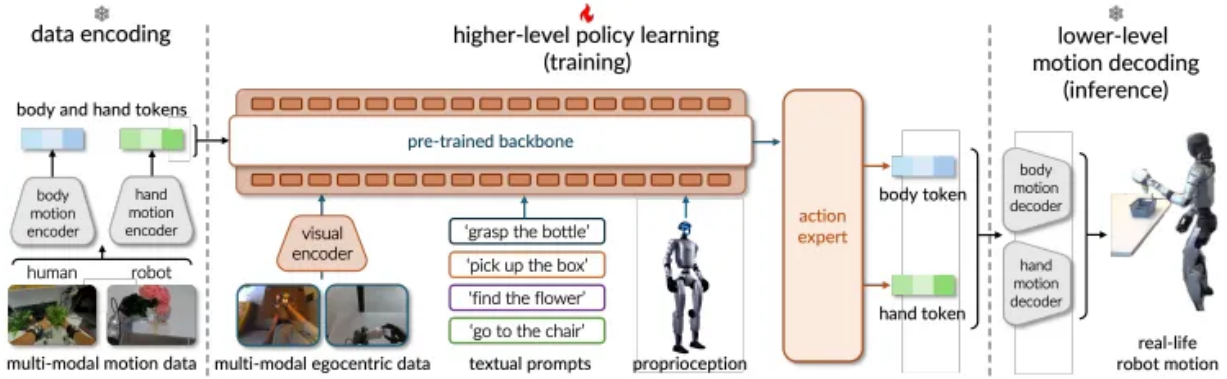
Semantic “common sense”, task decomposition, perception

↓

Coarse action head (low-level)

- Online generation of novel behaviors via primitive composition
- Few-shot learning via symbolic planning or online RL over primitives
- Dynamic stability, robustness, cross-embodiment transfer

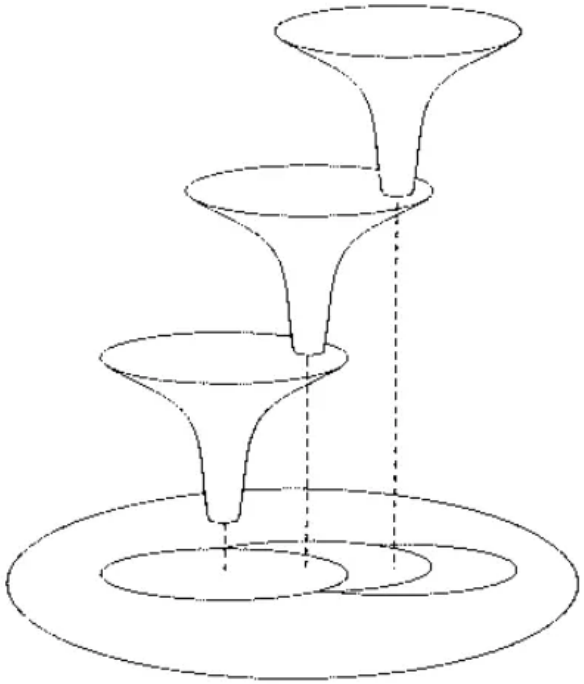
Joint actions to latent interfaces



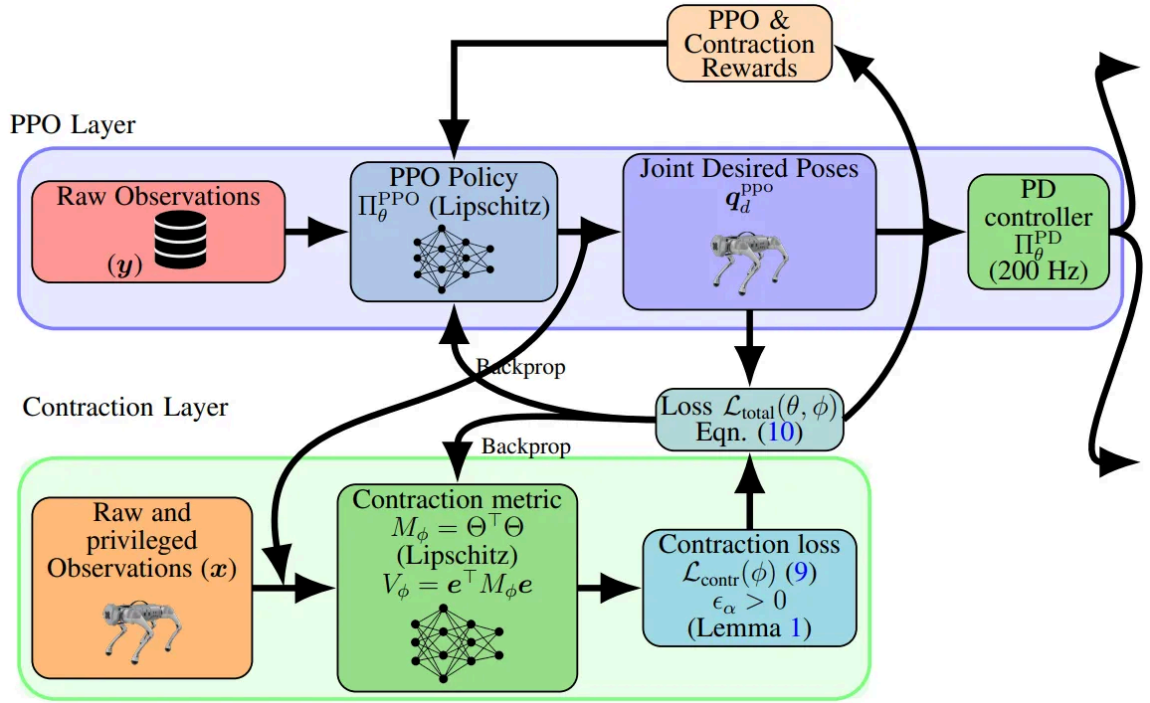
VioLA hierarchical connection of a “frozen encoder” with a latent (designed) interface (Albaba et al., 2026). Zero-shot locomotion success 100%, versus 16.7% for GR00T N1.7.

Open challenge: Aligning discrete/latent foundation model interfaces with formal composability when not trained end-to-end

Formal Composability of RL Primitives



Sequential composition formalism provides a preimage condition for composing controllers (Burridge et al., 1999)



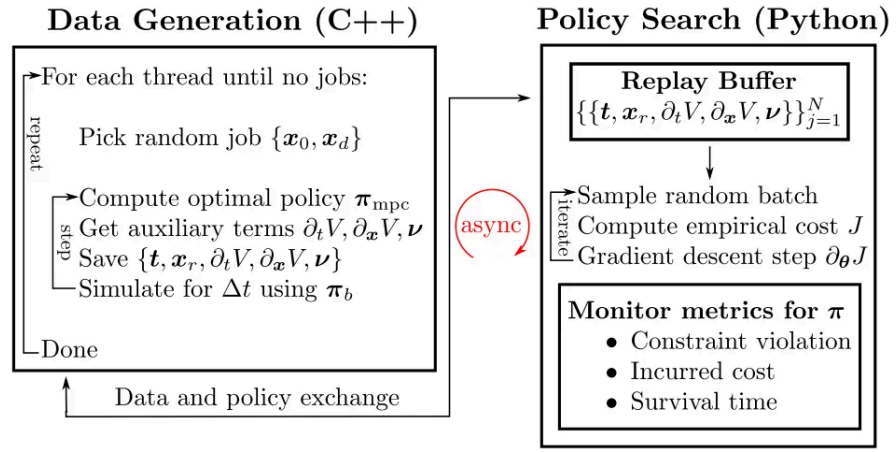
ContractionPPO adds a contraction loss during RL training for robustness (Zinage et al., 2026)

Idea: Add composability criteria to training loss for guaranteed online recombination

Generalization's Computational Footprint

	Optimization-based $\min J(x,u)\ \text{s.t. } g(x)=0$	**RL** $a=\pi_\theta(s)$
Online generalization	Strong	Weak
Computational requirement	High	Low

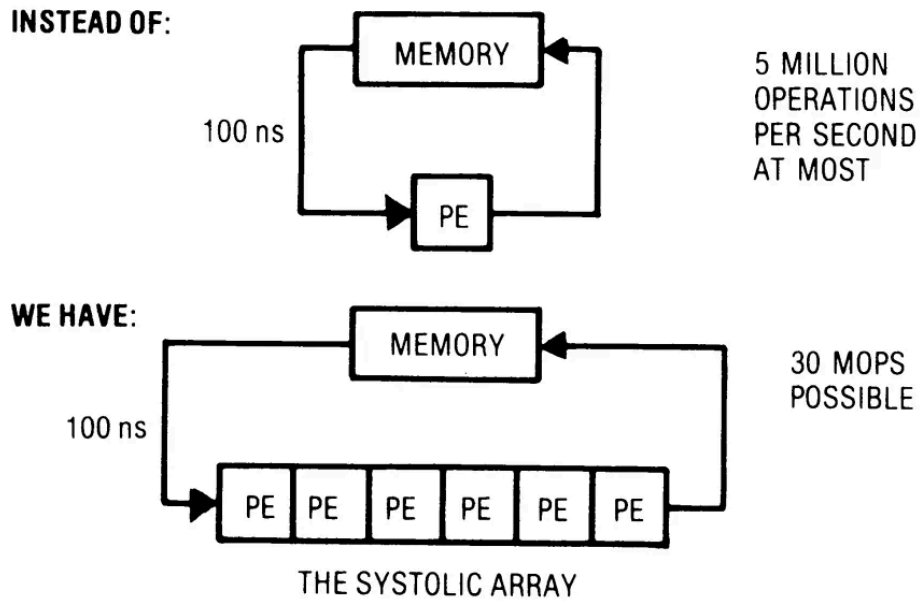
- Online MPC → NN distillation can be 10–70× (Khadiv et al., 2023) faster
- NN fails to generalize when OOD or into unseen contact modes where online MPC would re-solve (Reske et al., 2021)
- Model-based traj-opt inside RL training loop can be the primary bottleneck (Jenelten et al., 2024) (though not always (Bravo-Palacios et al., 2026)!)



MPC-Net "compiles" an MPC controller (Reske et al., 2021)

Regularity vs. Co-Design Potential

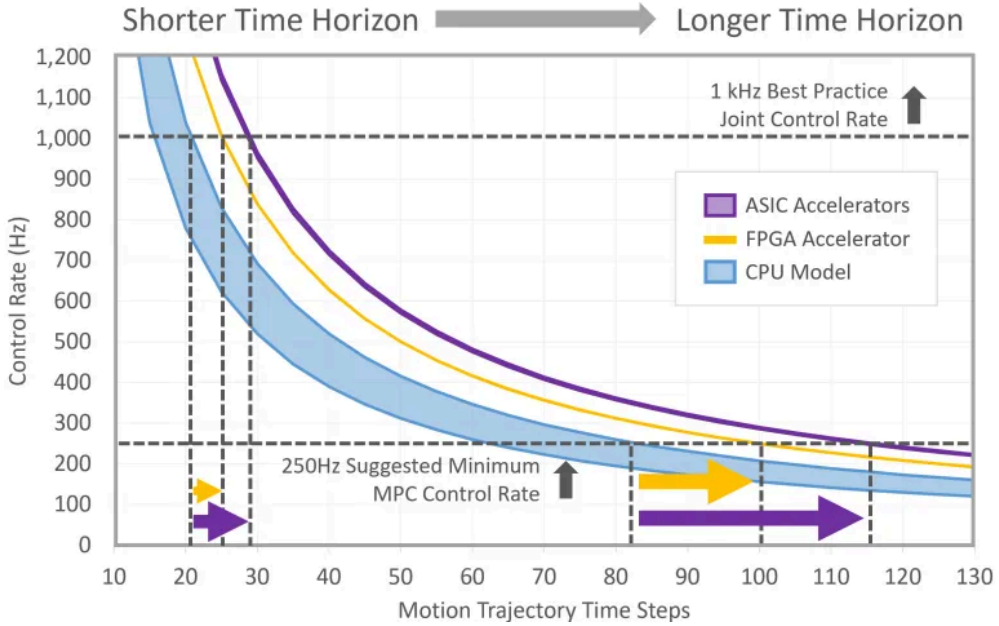
DNN regularity allows codesign



Systolic arrays make DNN computation faster and more efficient (Kung et al., 1979)

- TPU (Jouppi et al., 2017) – 30–80× perf/W
- Serial Riccati recursion resists acceleration (Plancher & Kuindersma, 2018)

ASICs: performant, but inflexible



Robotics codesign successes: Navion (Suleiman et al., 2019), Robomorphic ASIC (Neuman et al., 2021)

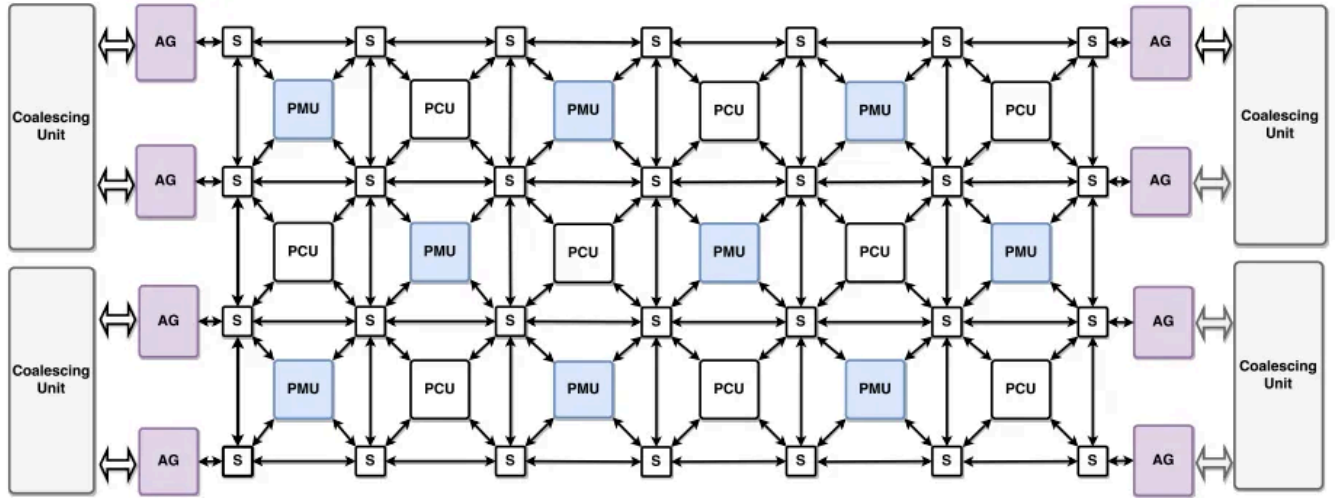
Huge perf/eff gains (e.g. 2 mW VIO; 40× dynamics grad), but inflexible

Computer Architecture Affects Throughput and Energy

Von Neumann vs. Dataflow

- **Stored-program bottleneck:** Data movement costs **10× (SRAM) to >150× (DRAM)** more energy than arithmetic (Horowitz, 2014)
- **MPC has low arithmetic intensity:** Dominated by memory movement (e.g., TinyMPC (Nguyen et al., 2024))
- **Data-flow and domain flow:** Organize computation around data movement (Omtzigt, 1990)

Flexible dataflow hardware is emerging



Plasticine (Prabhakar et al., 2017), SambaNova RDU compile whole compute graphs into spatial pipelines on-chip (up to 77× perf/W)

The opportunity for robotics:

- Systolic arrays (Kung et al., 1979) unlocked DNNs
- Domain-flow could unlock optimization-based control

Conclusion

What we want

- Robots that exhibit dynamical stability and robustness
- Can generate new behaviors online
- Generalize to new tasks and environments

What we have so far

- Ways to hand-craft composable dynamical primitives from templates
- Ways to build RL primitives that may not be composable

What we need to build

- Architectures that can flexibly and easily synthesize composable primitives
- Architectures that can compose them online to generate new behaviors

Thank you!

More information: avikde.github.io

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